(i) Let $a = \lfloor x \rfloor\,,\ b = x - \lfloor x \rfloor$

$$x = a + b, \qquad 0 \le b < 1$$

$$\begin{aligned} &\because n \in \mathbb{Z}^+ \\ &\therefore \lfloor x + n \rfloor = \lfloor a + b + n \rfloor \\ &\qquad\qquad = a + n \\ &\qquad\qquad = \lfloor x \rfloor + n \end{aligned}$$

(ii)(a)

$$\begin{aligned} f_n\left(x + \frac{1}{n}\right) &= \left\lfloor x + \frac{1}{n} \right\rfloor + \left\lfloor x + \frac{2}{n} \right\rfloor + \cdots + \lfloor x + 1 \rfloor - \lfloor nx + 1 \rfloor \\ &= \left\lfloor x + \frac{1}{n} \right\rfloor + \left\lfloor x + \frac{2}{n} \right\rfloor + \cdots + \left\lfloor x + \frac{n-1}{n} \right\rfloor + \lfloor x \rfloor + 1 - \lfloor nx \rfloor - 1 \\ &= \lfloor x \rfloor + \left\lfloor x + \frac{1}{n} \right\rfloor + \cdots + \left\lfloor x + \frac{n-1}{n} \right\rfloor - \lfloor nx \rfloor \\ &= f_n(x) \end{aligned}$$

$$\boxed{f_n\left(x + \frac{1}{n}\right) = f_n(x)}$$

(ii)(b) If $0 \le t < \dfrac{1}{n} \implies 0 \le t < 1$

For $0 \le t < \dfrac{1}{n} \implies 0 \le nt < 1 \implies \lfloor nt \rfloor = 0$

For $r = 0, 1, \ldots, n-1 \implies 0 \le t + \dfrac{r}{n} < \dfrac{1}{n} + \dfrac{r}{n} = \dfrac{r+1}{n} \le 1$

$\because t < \dfrac{1}{n} \quad \therefore t + \dfrac{r}{n} < 1 \implies \left\lfloor t + \dfrac{r}{n} \right\rfloor = 0$

$$\implies \boxed{f_n(t) = 0}$$

(ii)(c) Let $t = x - \dfrac{k}{n}$ and $0 \le x - \dfrac{k}{n} < \dfrac{1}{n} \implies x = t + \dfrac{k}{n}$

From part(ii)(a) $\implies f_n(x) = f_n\left(t + \dfrac{k}{n}\right) = f_n(t)$

$$\because f_n(t) = 0 \therefore \boxed{f_n(x) = 0}$$

(iii)(a) When $n = 2$ for part (ii)(c)

$$f_2(u) = \lfloor u \rfloor + \left\lfloor u + \frac{1}{2} \right\rfloor - \lfloor 2u \rfloor = 0$$

Let $u = \dfrac{x}{2} \quad \implies \left\lfloor \dfrac{x}{2} \right\rfloor + \left\lfloor \dfrac{x+1}{2} \right\rfloor - \lfloor x \rfloor = 0 \implies \boxed{\left\lfloor \dfrac{x}{2} \right\rfloor + \left\lfloor \dfrac{x+1}{2} \right\rfloor = \lfloor x \rfloor}$

(iii)(b) Let

$$S = \left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+2}{2^2} \right\rfloor + \cdots + \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor + \cdots$$

When $x = \dfrac{u}{2^k}$, from part (iii)(a),

$$\left\lfloor \frac{u}{2^{k+1}} \right\rfloor + \left\lfloor \frac{u+2^k}{2^{k+1}} \right\rfloor = \left\lfloor \frac{u}{2^k} \right\rfloor \implies \left\lfloor \frac{u+2^k}{2^{k+1}} \right\rfloor = \left\lfloor \frac{u}{2^k} \right\rfloor - \left\lfloor \frac{u}{2^{k+1}} \right\rfloor$$

$$\begin{aligned} \therefore S_N &= \sum_{k=0}^{N} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor \\ &= \sum_{k=0}^{N} \left(\left\lfloor \frac{x}{2^k} \right\rfloor - \left\lfloor \frac{x}{2^{k+1}} \right\rfloor \right) \\ &= \lfloor x \rfloor - \left\lfloor \frac{x}{2^{N+1}} \right\rfloor \end{aligned}$$

As $N \to \infty, \implies \left\lfloor \frac{x}{2^{N+1}} \right\rfloor = 0$

$$\implies \boxed{S = \lfloor x \rfloor}$$

This solution was submitted by **Camile**

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